PAISA PATH

A pixel life-sim where students learn money by living it: make choices, see consequences, try again.

1 · PROBLEM & OUR UNDERSTANDING

- **Who:** Grade 11–12 (Intermediate) students from underserved communities in Hyderabad; ~1,500 at launch through partner NGOs, designed to scale.
- **What:** four pillars: plan & budget, save & manage, understand investing, stay financially safe.
- **Financial literacy, as we read it:** not finance as a subject, but the knowledge, habits and confidence to make sensible decisions with *one's own* money: pocket money, scholarships, UPI, a first salary.
- **What makes it hard:** (1) relatability without jargon; (2) proving learning, not just engagement; (3) NGO delivery on shared, low-end phones in several languages; (4) minors' privacy; (5) the guardrail: educate, never recommend investments; simulations are assumptions.

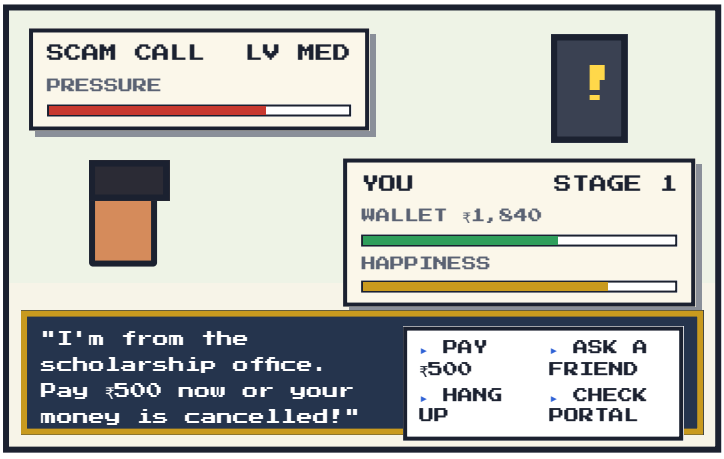
2 · IDEA / ABSTRACT

Paisa Path is a browser-based pixel RPG. A student creates an avatar, answers a short onboarding quiz, and lives a compressed life in three stages: **Intermediate student → first salary → earning more**. Each stage is a run of **task "battles"** across the four pillars (a scam call, a festival budget, a medical emergency) where the student chooses what to do and the game shows the consequence, sometimes much later. Mistakes are logged and come back in new disguises until learned. The game ends with an **epilogue at ~60** showing where their choices led, and a **personal report** (including their biggest mistake replayed the other way) sent to their NGO facilitator, with their guardian copied.

3 · PLAYER JOURNEY (PIPELINE)

01	02	03	04	05	06
Onboard	**Persona + economy**	**Difficulty quiz**	**Three life stages**	**Epilogue**	**Report**
Language, avatar + chosen name, pocket-money band, a few life questions	Persona sets how tasks are framed. Prices scale to the student's pocket money	Places the student in a bucket: simple · medium · hard. Repeated at the end	School → first salary → earning more. Task battles, random events, mini-games	Age ~60: what savings and investments became (labelled as assumptions)	Score + biggest-mistake "what if", emailed to the NGO facilitator, guardian CC'd

EVERY TASK IS A "MONEY BATTLE"



Concept mock-up with original art. Opponent = the situation; its bar = pressure. Your bars = wallet and happiness. 2–4 choices per turn.

WHAT HAPPENS AFTER A CHOICE

- **Instant effect:** wallet, savings and happiness bars move; a 1–2 line lesson names the concept (taught *after* the choice, never as a lecture first).
- **Weak choice → detour:** extra consequence tasks, then the path rejoins.
- **Critical failure → restart the task:** falling for a scam, going broke, or happiness hitting zero.
- **Delayed effects:** some choices schedule a later event (an EMI deducted every month; a friend's unpaid loan resurfacing).
- **Mistake bank:** each wrong concept returns later in a new disguise; it's cleared only when answered right.
- **Info button:** current status at any point. **Replay library:** re-watch the short explainer animation for any past task.
- **Random mini-games:** skippable and rewarding, but locked while a status bar sits below its threshold.

HOW WE MEET THE SUCCESS MEASURES

SIMPLE One decision per screen; lessons in 1–2 lines; no jargon	**RELATABLE** Hyderabad places, everyday rupee situations, prices scaled to the student's own pocket money	**PERSONA-LED** Persona framing for every task, plus difficulty buckets from the quiz	**LANGUAGE-READY** One string file per language; Bhashini-assisted translation; audio narration
GAME-BASED Learning measured through in-game choices, plus a before/after quiz	**INTUITIVE UI** Tap-to-travel map, battle screen, info button, no sign-up form	**NGO-INTEGRATED** Batch join codes, facilitator dashboard, reports emailed to facilitators	**SAFE SANDBOX** Fake money, detours and restarts: mistakes cost nothing real

4 · ARCHITECTURE

CLIENT · PHASER 3

- Runs in any phone browser; offline-capable (PWA)
- Scenes: onboarding, life map, task battle, mini-games, info panel, replay library, end report
- Tap-to-travel town map built from Hyderabad landmarks (Charminar, Durgam Cheruvu, Gachibowli)

GAME ENGINE · PLAIN JS, UNIT-TESTED

- **State store:** wallet, savings, happiness, stage, flags
- **Economy scaler:** real prices × (in-game start ÷ pocket money), so items cost the same share of budget as in real life
- **Task selector:** pillar × difficulty bucket, then persona framing
- **Consequence scheduler:** delayed effects
- **Mistake bank + scoring:** NCFE-aligned (knowledge, behaviour, attitude) + a safety score
- **History log:** every decision + its alternatives, for the "what if"

CONTENT · JSON

- Task buckets per pillar × difficulty (simple/medium/hard), each with a framing per persona
- Random events, mini-games, explainer animations
- One string file per language
- Editable by non-coders, and by NGOs adapting it to new cities

Translation
Indian-language strings via Bhashini translation APIs, then human-checked

NGO dashboard
Batch join codes, progress per centre, aggregate before/after scores

Report service
End report as PDF, emailed to the facilitator with the guardian CC'd

Storage
Local-first save; syncs when online. Nickname only, no student PII

Client ⇄ Engine ⇄ Content · Engine → Report service → NGO facilitator + guardian · Engine → anonymised aggregates → NGO dashboard / sponsor

KEY DESIGN DECISIONS

MECHANIC	HOW IT WORKS
Persona	Assigned from life questions (incl. a pocket-money band); shown to the student, who can switch. Changes only the **framing and analogies** of each task, not its difficulty.
Difficulty	Onboarding quiz places the student in simple / medium / hard; tasks are drawn from that bucket. Same quiz at the end for before/after.
Economy scaling	Everyone starts with the same in-game amount; prices are scaled to the student's own pocket money, so a cricket bat "feels" as expensive as it does in their real life.
Life stages	Stage 1 Intermediate student → Stage 2 first salary → Stage 3 earning more. Investing grows naturally across stages; epilogue at ~60.
Biggest-mistake "what if"	From the decision log, the choice with the largest negative impact is replayed with the alternative, using the same random seed, and shown side by side.
Guardrail	Fictional assets only, never real products; every projection carries "assumes X% a year; real returns vary and can be negative".

5 · PLANNED TESTING METHODOLOGY

Learning gain
Same difficulty quiz before and after; gain per pillar. In-game choices scored on the NCFE framework (knowledge ≥ 6/8, behaviour ≥ 6/9, attitude ≥ 3/5), with India's 27% adult baseline as reference.

Usability
Unguided play tests (peers as proxies during the hackathon, target-age students in a pilot): time to finish the first task with no help, help requests, drop-off points.

Engine correctness
Unit tests for the economy scaler, consequence scheduler, mistake bank, scoring and "what if" replay on seeded game histories. Schema checks that every task has a difficulty tag, all persona framings and valid effects.

Low-end devices
Chrome DevTools throttling (slow 3G, 4× CPU slowdown) on a budget-Android profile; offline play-through; load-time target under 5 s.

Language
Automated check that every string key exists in each language; screenshot review per language.

Report & privacy
Reports generated for seeded histories and sent to sandbox inboxes. Content checklist: no product recommendations, all projections labelled. No PII beyond a nickname.